Computer Vision Practical Report:
LAPLACIAN MASK WITH POSITIVE CENTRE COEFFICIENT

1. AIM

To sharpen an image using a Laplacian filter with a positive center coefficient in OpenCV, display the original and sharpened images, and save the output.

2. PROCEDURE

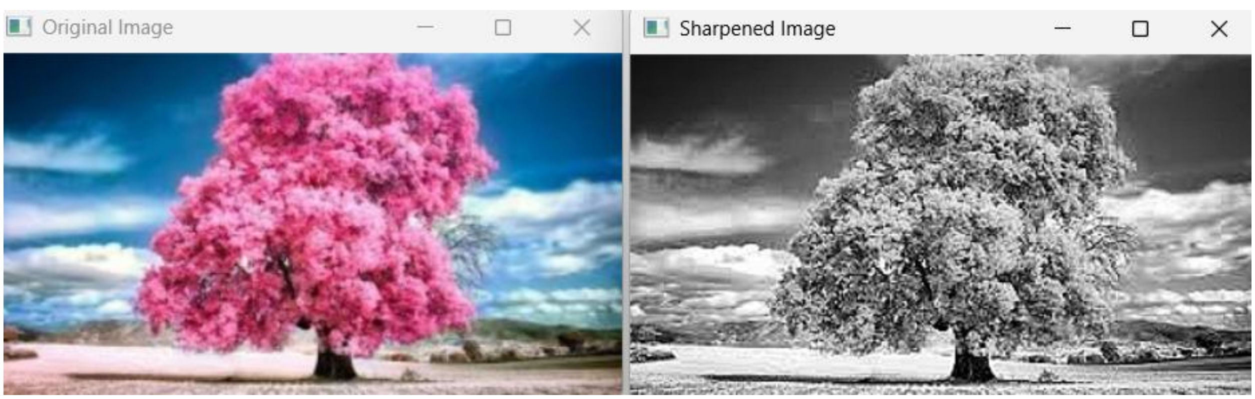
- Load the image and convert it to grayscale.
- Define a 3×3 Laplacian kernel with a positive center value (e.g., center = $+4$ or $+8$, edges = -1).
- Filter the image using cv2.filter2D() to obtain the Laplacian response.
- Add the Laplacian response to the original image to obtain the sharpened output.
- Display the original and sharpened images side-by-side using Matplotlib, and save the result using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import numpy as np

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E22-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Laplacian sharpening mask with positive center
kernel = np.array([
    [0, -1, 0],
    [-1, 5, -1],
    [0, -1, 0]
```

```
], dtype=np.float32)

# Apply sharpening mask
sharpened = cv2.filter2D(gray, -1, kernel)

cv2.imshow("Original Image", image)
cv2.imshow("Sharpened Image", sharpened)

cv2.imwrite("laplacian_positive_sharpened.jpg", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.